University of Alberta
STAT 441 Final Exam
Winter Term 2023

Name (print): _____

Signature: _____

Student ID: _____

Date of Exam:	April 25, 2023
Time Period:	14:00 pm - 17:00 pm
Duration of Exam:	180 minutes
Number of Exam Pages:	18 (this cover page included)
Exam Type:	Closed Book
Additional Materials Allowed:	A Calculator

Instructions:
[1] Read all questions before you start.
[2] Answer easy parts first! Understand the questions first! Make it short and accurate!
[3] The last five pages are the `R` code and `R` output.
[4] Raise you hand if you have any questions.
[5] Please make sure you hand in back all your 18 exam pages.

Marking Scheme:

Question	Score	Question	Score
Question 1 (6)		Question 2 (6)	
Question 3 (6)		Question 4 (6)	
Question 5 (8)		Question 6 (8)	
Total (40)			

Question 1 (6 points)

[1] (3 points) In linear discriminant analysis, suppose that the response y has K classes and the covariate x is of one dimension. Starting from the Bayes Theorem and the necessary assumptions, derive the linear discriminant score $\delta_k(x)$ for $k = 1, \ldots, K$. In particular, if $K = 2$, what is the decision boundary assuming equal priors?

[2] (3 points) In linear regression, how to do variable selection using LASSO? Explain why it can be used to do variable selection and describe how to use K-fold cross validation to choose the shrinkage tuning parameter.

Question 2 (6 points)

[1] (4 points) In a single hidden layer back-propagation network, we have input $X = (X_1, \cdots, X_p)^T$ and response $Y = (Y_1, \cdots, Y_K)^T$ and the following model

$$Z_m = \sigma(\alpha_{0m} + \alpha_m^T X),\ m = 1, \cdots, M;\ T_k = \beta_{0k} + \beta_k^T Z;\ f_k(X) = g_k(T),\ k = 1, \cdots, K, \tag{1}$$

where $Z = (Z_1, \cdots, Z_M)^T$, and $T = (T_1, \cdots, T_K)^T$. The activation function $\sigma(v)$ is usually chosen to be the sigmoid function $\sigma(v) = 1/(1+e^{-v})$. Suppose the risk function is $R(\theta) = \sum_{i=1}^n R_i$. The generic approach to minimizing $R(\theta)$ is by gradient descent, called back-propagation in this setting. Let $z_{mi} = \sigma(\alpha_0 m + \alpha_m^T x_i)$ and $z_i = (z_{1i}, \cdots, z_{Mi})$. The derivatives of $R_i(\theta)$ can be rewritten as

$$\frac{\partial R_i}{\partial \beta_{km}} = \delta_{ki} z_{mi},\ \frac{\partial R_i}{\partial \alpha_{kl}} = s_{mi} x_{il}, \tag{2}$$

where the quantities δ_{ki} and s_{mi} are errors from the current model at the output and hidden layer units, respectively. Given the derivatives, a gradient decent update at $(r+1)$ iteration has the form

$$\beta_{km}^{(r+1)} = \beta_{km}^{(r)} - \gamma_r \sum_{i=1}^n \frac{\partial R_i}{\partial \beta_{km}^{(r)}},\ \alpha_{ml}^{(r+1)} = \alpha_{ml}^{(r)} - \gamma_r \sum_{i=1}^n \frac{\partial R_i}{\partial \alpha_{kl}^{(r)}}, \tag{3}$$

where γ_m is the learning rate. Now consider a multiclass classification problem, we usually choose g as the softmax function and risk R as the cross entropy. In this case what are the forms of the errors of δ_{ki} and s_{mi}? Derive the back-propagation equation and describe the back-propagation algorithm.

[2] (2 points) Describe what are Factor Analysis and Principal Component Analysis? Both methods are trying to interpret the covariance matrix, what is the difference?

Question 3 (6 points)

In this question, we will study the Boston House Prices using the `Boston` data, which is available from the `MASS` package in `R`. The methods we are focusing are tree-based methods. the goal is to predict `medv` using all the other variables. All the necessary `R` outputs are listed in the Appendix.

[1] (3 points) In the `Boston` data, there are 13 covariates available to predict `medv`. What is the appropriate number of variables to choose in each step for a new bootstrapped sample if we use the method of random forest? What are the first five important variable judging by the amount of RSS reduced?

[2] (3 points) What is the difference between boosting and bagging? Given a new observation with

```
   crim zn indus chas   nox    rm  age  dis rad tax ptratio black lstat medv
0.00632 18  2.31    0 0.538 6.575 65.2 4.09   1 296    15.3 396.9  4.98   24
```

what is the predicted `medv` by boosting?

Question 4 (6 points)

In this problem, we will use Clustering Analysis (CA) to study the `USArrests` data set where the covariates are

`Murder, Assault, UrbanPop, Rape.`

To simplify the question, we randomly choose only 6 states,

`Georgia, Louisiana, Washington, Oklahoma, Tennessee, California`

All the necessary `R` outputs are listed in the Appendix.

[1] (3 points) What does Clustering Analysis (CA) do? In hierarchically clustering, the clustering results will be affected by the linkage we choose to merge two clusters based on dissimilarity measures. What are the definitions of single, complete, average and centroid linkages?

[2] (3 points) Picking up any linkage from the above four type of linkages, describe the hierarchically clustering procedure to form K clusters. Choosing any two linkages to build the dendrograms and then form $K = 2$ clusters based on the dissimilarity measure of Euclidean distance, which states(observations) are in each cluster? see the Appendix for the paired Euclidean distance matrix. (For instance, the dissimilarity between Georgia and Louisiana is 38.69057, and the dissimilarity between Washington and Tennessee is 46.15333.)

Question 5 (8 points)

In this problem, we will apply Canonical Correlation Analysis (CCA) on the `crime` data set about a study of poverty, crime and deterrence, where summary crime statistics in various states were reported for the years 1970 and 1973. A portion of their sample correlation matrix is

$$\mathbf{R} = \left[\begin{array}{c:c} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \hdashline \mathbf{R}_{21} & \mathbf{R}_{22} \end{array}\right] = \left[\begin{array}{cc:cc} 1.0 & .615 & -.111 & -.266 \\ .615 & 1.0 & -.195 & -.085 \\ \hdashline -.111 & -.195 & 1.0 & -.269 \\ -.266 & -.085 & -.269 & 1.0 \end{array}\right]$$

The variables are

$$\begin{aligned} X_1^{(1)} &= \text{1973 nonprimary homicides} \\ X_2^{(1)} &= \text{1973 primary homicides (homicides involving family or acquaintances)} \\ X_1^{(2)} &= \text{1970 severity of punishment (median months served)} \\ X_2^{(2)} &= \text{1970 certainty of punishment (number of admissions to prison divided by number of homicides)} \end{aligned}$$

All the necessary `R` outputs are listed in the Appendix.

[1] (4 points) What does Canonical Correlation Analysis (CCA) do? Describe the conditions that the first two pair of canonical variables and their correlations should satisfy. Illustrate how to use the Singular Value Decomposition (SVD) of covariance matrix $\mathbf{\Sigma}_{(p+q)\times(p+q)}$, where p is the number of variables in the first set and q is the number of variables in the second set, to obtain the first two pair of canonical variables and their correlations.

[2] (4 points) What are the first two pair of canonical variables and their correlations for the `crime` data set (Rounding to four decimal places)? How to interpret the first pair of canonical variable? See the necessary `R` outputs are listed in the Appendix.

Question 6 (8 points)

In this problem, we will use Factor Analysis (FA) to study the `Sales` data set. A firm is attempting to evaluate the quality of its sales staff and is trying to find an examination or series of tests that may reveal the potential for good performance in sales. The firm has selected a random sample of 50 sales people and has evaluated each on 3 measures of performance: growth of sales, profitability of sales, and new-account sales. These measures have been converted to a scale, on which 100 indicates 'average' performance. Each of the 50 individuals took each of 4 tests, which purported to measure creativity, mechanical reasoning, abstract reasoning, and mathematical ability, respectively. The $p = 7$ variables are, respectively,

`growth, profitability, new, sales, creativity, mechanical, abstract, math.`

All the necessary `R` outputs are listed in the Appendix.

[1] (4 points) The orthogonal factor model has the form of

$$\mathbf{x}_{p\times 1} = \boldsymbol{\mu} + \mathbf{L}_{p\times m}\mathbf{f}_{m\times 1} + \boldsymbol{\varepsilon},$$

where the elements $F_1, ..., F_m$ of $\mathbf{f}$ are r.v.s called common factors and $\mathbf{L}$ is a non-random matrix of factor loadings with rank $m \leq p$. The mean-zero random errors ε_i are uncorrelated with each other and also with the F_j; we assume as well that

$$E[\mathbf{f}] = \mathbf{0},\ \text{COV}[\mathbf{f}] = \mathbf{I}_m,\ \text{COV}[\boldsymbol{\varepsilon}] = \boldsymbol{\Psi} = diag(\psi_1, ..., \psi_p).$$

Illustrate how to use PCA to obtain $\boldsymbol{\mu}$, $\boldsymbol{L}$ and $\boldsymbol{\Psi}$; and then if $\boldsymbol{\mu}$, $\boldsymbol{L}$ and $\boldsymbol{\Psi}$ are known, how to obtain the estimate of $\boldsymbol{f}$ using the weighted least square method.

[2] (4 points) Now we further assume that the data was standardized. If we want the PCAs to explain at least 95% total variance, how many factors m we should choose? Find the corresponding $\boldsymbol{L}$ and $\boldsymbol{\Psi}$ (Rounding to three decimal places). Suppose a new salesperson, selected at random, obtains the (unstandardized) test scores $\mathbf{x}^T = (110, 98, 105, 15, 18, 12, 35)$. Calculate the salesperson's factor scores.

Appendix: R code and output

```
> #################### STAT 441 final exam #################
> ################### JHX ################
> ##############################################################
>
> rm(list = ls())
> ###################### Question 3 #################
> # Load the tree package;
> library(tree)
>
> set.seed(1)
> tree.boston=tree(medv~.,Boston)
> par(mfrow=c(1,2))
> prune.boston=prune.tree(tree.boston,best=4)
> plot(prune.boston)
> text(prune.boston,pretty=0)
> prune.boston=prune.tree(tree.boston,best=5)
> plot(prune.boston)
> text(prune.boston,pretty=0)
>
> newdata = Boston[1,]
> newdata
     crim zn indus chas   nox    rm  age  dis rad tax ptratio black lstat medv
1 0.00632 18  2.31    0 0.538 6.575 65.2 4.09   1 296    15.3 396.9  4.98   24
> prune.boston=prune.tree(tree.boston,best=6)
> predict(prune.boston,newdata=newdata)
       1
27.42727
>
> n = dim(Boston)[1]
> nn = 1000
> pred.y = c(1:nn)
> set.seed(1)
> for (i in 1:nn){
+  sam = sample(n,n,rep=T)
+  new.Boston = Boston[sam,]
+  tree.new.boston=tree(medv~.,new.Boston)
+  prune.new.boston=prune.tree(tree.new.boston,best=5)
+  pred.y[i] = predict(prune.new.boston,newdata=newdata)
+ }
>
> quantile(pred.y, probs = c(0.025,0.05,0.5,0.95,0.975))
    2.5%       5%      50%      95%    97.5%
22.02772 22.19843 25.02916 32.80858 33.44481
> mean(pred.y)
[1] 25.65466
>
> library(randomForest)
> set.seed(1)
> rf.boston=randomForest(medv~.,data=Boston,mtry=4,importance=TRUE)
> importance(rf.boston)
```

```
          %IncMSE IncNodePurity
crim    17.239248     2486.0422
zn       5.276511      284.0270
indus   11.774606     2838.7832
chas     4.323979      231.3077
nox     19.932378     2707.7264
rm      37.148007    12416.3171
age     12.484465     1076.3452
dis     19.906481     2540.4924
rad      6.752833      368.4620
tax     13.718877     1309.7731
ptratio 16.013591     2619.5196
black    9.374145      762.2800
lstat   30.769442    12283.5197
> rf.boston=randomForest(medv~.,data=Boston,mtry=5,importance=TRUE)
> importance(rf.boston)
          %IncMSE IncNodePurity
crim    15.816222     2226.2115
zn       3.704310      206.1187
indus   11.147850     2280.1220
chas     1.810362      132.6960
nox     18.904639     2472.0552
rm      40.578996    13361.5749
age     13.714080      963.9306
dis     20.190690     2378.6972
rad      7.656916      311.5129
tax     10.462477     1053.9516
ptratio 16.629607     2265.7346
black    9.894316      621.9091
lstat   33.547287    13703.0430
> rf.boston=randomForest(medv~.,data=Boston,mtry=6,importance=TRUE)
> importance(rf.boston)
          %IncMSE IncNodePurity
crim    15.842303     2048.5124
zn       2.416669      123.6728
indus   10.930066     1909.8736
chas     2.855304      118.3181
nox     21.342840     2404.2278
rm      42.538916    13693.4494
age     14.186221      918.1509
dis     19.463897     2518.4485
rad      6.631365      234.6340
tax     11.263114      827.9384
ptratio 15.638923     2246.4345
black   10.018520      564.0895
lstat   34.912686    14804.9755
>
> library(gbm)
> set.seed(1)
> boost.boston=gbm(medv~.,Boston,distribution="gaussian",n.trees=1000,interaction.depth=3)
> predict(boost.boston,newdata=newdata,n.trees=1000)
[1] 25.35321
```

```
>
> rm(list = ls())
> ###################### Question 4 ##############################
> library(ISLR)
>
> data(USArrests)
> names(USArrests)
[1] "Murder"   "Assault"  "UrbanPop" "Rape"
>
> n = dim(USArrests)[1]
> set.seed(11)
> sam = sample(n,6,rep=F)
> USArrests01 = USArrests[sam,]
> dist(USArrests01, diag = T)
             Georgia Louisiana Washington  Oklahoma Tennessee California
Georgia      0.00000
Louisiana   38.69057   0.00000
Washington  68.59096 104.93312    0.00000
Oklahoma    61.75986  98.43922   10.30534   0.00000
Tennessee   23.42755  61.61923   46.15333  39.25774   0.00000
California  73.99730  41.63556  133.10657 128.77935  94.72766    0.00000
>
> rm(list = ls())
> ###################### Question 5 ##############################
>
>
> S11 = cbind(c(1,0.615), c(0.615,1))
> S12 = cbind(c(-0.111,-0.195), c(-0.266,-0.085))
> S21 = t(S12)
> S22 = cbind(c(1,-0.269), c(-0.269, 1))
>
> p = nrow(S11)
> q = nrow(S22)
> s1 = eigen(S11)
> rootS11 = s1$vectors%*%diag(sqrt(s1$values))%*%t(s1$vectors)
> solve(rootS11)
           [,1]       [,2]
[1,]  1.1992677 -0.4123782
[2,] -0.4123782  1.1992677
>
> s2 = eigen(S22)
> rootS22 = s2$vectors%*%diag(sqrt(s2$values))%*%t(s2$vectors)
> solve(rootS22)
          [,1]      [,2]
[1,] 1.0286584 0.1409523
[2,] 0.1409523 1.0286584
>
> S12tilde = solve(rootS11, S12)%*%solve(rootS22)
>
> s = svd(S12tilde, nu = p, nv = q)
> G = s$u
> G
```

```
           [,1]       [,2]
[1,] -0.9463168 -0.3232407
[2,] -0.3232407  0.9463168
> H = s$v
> H
          [,1]       [,2]
[1,] 0.4634266 -0.8861353
[2,] 0.8861353  0.4634266
> R = cbind(diag(s$d),matrix(nrow=p, data = rep(0, p*(q-p))))
> R
          [,1]      [,2]
[1,] 0.3266219 0.0000000
[2,] 0.0000000 0.1710696
>
> rm(list = ls())
> ##################### Question 6 ##############################
>
> tr = function(M) sum(diag(M))
>
> Sales = read.table("/Volumes/MHD2/Study/2023/teach/STAT441/Final/Sales.DAT")
> X = as.matrix(Sales)
> colnames(X) = c("growth", "profitability", "new sales",
+                 "creativity", "mechanical", "abstract", "math")
> R = cor(X)
> R
                 growth profitability new sales creativity mechanical  abstract      math
growth        1.0000000     0.9260758 0.8840023  0.5720363  0.7080738 0.6744073 0.9273116
profitability 0.9260758     1.0000000 0.8425232  0.5415080  0.7459097 0.4653880 0.9442960
new sales     0.8840023     0.8425232 1.0000000  0.7003630  0.6374712 0.6410886 0.8525682
creativity    0.5720363     0.5415080 0.7003630  1.0000000  0.5907360 0.1469074 0.4126395
mechanical    0.7080738     0.7459097 0.6374712  0.5907360  1.0000000 0.3859502 0.5745533
abstract      0.6744073     0.4653880 0.6410886  0.1469074  0.3859502 1.0000000 0.5663721
math          0.9273116     0.9442960 0.8525682  0.4126395  0.5745533 0.5663721 1.0000000
>
> fit.pc <- princomp(X, cor=TRUE)
> summary(fit.pc)
Importance of components:
                          Comp.1    Comp.2     Comp.3     Comp.4    Comp.5      Comp.6      Comp.7
Standard deviation     2.2437909 0.9661864 0.70563429 0.64903427 0.2846760 0.142620593 0.106488339
Proportion of Variance 0.7192283 0.1333594 0.07113139 0.06017793 0.0115772 0.002905805 0.001619967
Cumulative Proportion  0.7192283 0.8525877 0.92371910 0.98389702 0.9954742 0.998380033 1.000000000
> V = fit.pc$loadings
> L = V%*%diag(fit.pc$sdev)
> L
                  [,1]         [,2]         [,3]         [,4]          [,5]        [,6]        [,7]
growth        0.9730692  0.107975602  0.053267303 -0.027501814 -1.800559e-01  0.04800556  0.05620724
profitability 0.9428714 -0.028297179  0.312228320  0.006978773  3.365135e-05 -0.11200599  0.01059381
new sales     0.9447504 -0.008890823 -0.144082982 -0.210889654  1.995654e-01  0.02236453  0.04250633
creativity    0.6603168 -0.645814259 -0.318588471 -0.196470515 -7.430272e-02 -0.01628313 -0.03194220
mechanical    0.7832895 -0.284971246 -0.004178606  0.549474729  4.960863e-02  0.02808330 -0.00770032
abstract      0.6488301  0.620656841 -0.426047603  0.099739764 -2.475503e-02 -0.03369568 -0.02432657
math          0.9141297  0.193592498  0.306273206 -0.159670997  1.411499e-02  0.05292801 -0.06775038
```

```
>
> L2.pc = L[,1:2]
> psi2.pc = diag(R) - diag(L2.pc%*%t(L2.pc))
> psi2.pc
       growth profitability     new sales    creativity    mechanical      abstract          math
   0.04147769    0.11019280    0.10736764    0.14690562    0.30524892    0.19380457    0.12688883
>
> L3.pc = L[,1:3]
> psi3.pc = diag(R) - diag(L3.pc%*%t(L3.pc))
> psi3.pc
       growth profitability     new sales    creativity    mechanical      abstract          math
   0.03864028    0.01270628    0.08660774    0.04540700    0.30523146    0.01228801    0.03308555
>
> L4.pc = L[,1:4]
> psi4.pc = diag(R) - diag(L4.pc%*%t(L4.pc))
> psi4.pc
       growth profitability     new sales    creativity    mechanical      abstract          math
  0.037883930   0.012657572   0.042133292   0.006806339   0.003308983   0.002339992   0.007590721
>
> xbar = colMeans(X)
> stdev = sqrt(diag(cov(X)))
> x = c(110, 98, 105, 15, 18, 12, 35)
> x0 = (x-xbar)/stdev
> f = t(V)%*%x0/fit.pc$sdev
> f
             [,1]
Comp.1  0.6999133
Comp.2 -0.2585665
Comp.3 -1.3974812
Comp.4  0.6505677
Comp.5 -2.5419175
Comp.6  9.7625016
Comp.7  0.6115964
```